

HIGHLAND HOME 2 S, AL, USA

WMO: 722266

Lat: 31.9155N

Lon: 86.3115W

Elev: 614

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 73802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.0	(c) 28.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.8	(c) 93.4	(d) N/A	(e) 91.4	(f) N/A	(g) 89.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) 17.6	(f) 95.4	(g) 4.6	(h) 3.5	(i) 14.3	(j) 97.9	(k) 11.6	(l) 99.9	(m) 9.0	(n) 101.8	(o) 5.7	(p) 104.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	47.0	52.3	58.2	65.1	72.2	78.4	79.8	79.6	76.4	67.2	56.1	51.4		
	DBStd	13.57	10.64	9.93	8.84	6.00	5.43	3.43	2.88	3.09	4.81	7.27	8.26	9.66		
	HDD50	450	184	87	34	1	0	0	0	0	0	2	35	107		
	HDD65	2025	559	363	240	72	15	0	0	0	2	63	282	429		
	CDD50	6061	93	153	287	453	687	852	923	918	791	534	219	151		
	CDD65	2162	3	9	28	74	237	402	458	453	343	131	16	8		
	CDH74	17552	1	25	137	507	1757	3580	4024	3825	2747	878	61	10		
	CDH80	6573	0	0	8	55	506	1468	1672	1593	1057	208	6	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	55.2	5.0	5.2	6.6	4.4	4.0	4.9	5.0	4.6	4.1	2.9	4.4	4.5		
	PrecMax	87.8	10.5	12.8	14.1	14.0	9.6	12.9	11.0	9.5	13.2	12.1	11.9	9.8		
	PrecMin	42.2	0.9	1.2	0.5	0.4	0.1	1.2	1.2	0.9	0.5	0.0	0.7	1.6		
	PrecStd	9.3	2.0	2.7	3.2	2.9	2.2	3.0	2.4	1.9	3.0	2.4	2.4	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	77.8	81.4	83.8	90.3	95.4	94.9	95.8	94.7	89.0	79.6	75.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	69.4	73.3	77.8	81.8	87.6	92.6	92.3	92.9	91.3	85.1	75.2	71.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	66.2	70.0	74.9	79.6	85.3	90.1	90.2	90.5	89.0	82.1	72.3	69.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	62.8	67.0	71.5	77.2	82.9	87.9	88.4	88.2	86.5	79.1	69.7	66.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	18.5	18.9	19.7	18.2	17.3	16.8	17.1	17.7	19.3	19.7	16.6		
		MCDBR	17.9	18.5	19.9	21.3	20.4	20.2	18.9	20.1	20.7	20.5	20.0	17.2		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.327	0.352	0.379	0.426	0.465	0.521	0.505	0.425	0.366	0.340	0.322		
	taud		2.537	2.517	2.445	2.367	2.273	2.220	2.069	2.112	2.318	2.462	2.477	2.539		
	Ebn at Noon		288	296	295	289	275	263	247	250	268	278	277	281		
	Edh at Noon		27	30	35	39	43	45	52	50	39	31	28	25		
(44) All-Sky Solar Radiation	RadAvg		908	1088	1465	1813	1989	1945	1889	1754	1570	1347	1026	783		
	RadStd		61	135	129	142	137	153	122	106	125	162	93	85		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+180

Nomenclature: See separate page